The documentation process conversion measures necessary to comply with this revision shall be completed by 11 May 1998

INCH-POUND

MIL-PRF-19500/607A
11 February 1998
SUPERSEDING
MIL-S-19500/607
1 September 1992

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTORS, N-CHANNEL AND P-CHANNEL, SILICON TYPE 2N7337 JANTX, JANTXV, JANS, AND JANC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for N-channel, and P-channel, enhancement-mode, MOSFET, power transistor. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500, with avalanche energy ratings (E_{AS} and E_{AR}) and maximum avalanche current (I_{AR}). One level of product assurance for die (element evaluation).

1.2 Physical dimensions. See figure 1 (MO-O36AB) (14 pin dip), see figure 2 for JANC die dimensions.

1.3 Maximum ratings (unless otherwise specified , T_A = +25°C).

Type	P_T 1/ T_C = +25°C	V_{GS}	I_{D1} 2/ T_C = +25°C		I_{D2} 2/ T_C = +100°C		I_S		Min $V_{(BR)DSS}$ V_{GS} = 0 V I_D = 1.0 mA	Min $V_{(BR)DSS}$ V_{GS} = 0 V I_D = -1.0 mA
	W	V dc	A dc		A dc		A dc		V dc	
			N-channel	P-channel	N-channel	P-channel	N-channel	P-channel	N-channel	P-channel
2N7337	1.4	± 20	1.0	1.0	0.6	0.6	1.0	1.0	100	100

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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1.3 Maximum ratings (unless otherwise specified , T_A = +25°C) Continued.

E_{AS}	E_{AR}	I_{AR} 2/		I_{DM} 3/		T_{OP} and T_{STG}	Max $R_{DS(ON)}$ 1/ V_{GS} = 10 V dc, I_D = I_{D2}				$R_{\theta JA1}$ maximum (1 die)	$R_{\theta JA2}$ maximum (4 die)
							ohms T_J = +25°C		ohms T_J = +150°C			
mJ	mJ	A		A(pk)		°C					°C/W	°C/W
		N-CH	P-CH	N-CH	P-CH		N-CH	P-CH	N-CH	P-CH		
75	.14	1.0	1.0	4.0	4.0	-55 to +150	0.7	0.7	1.4	1.4	90	50

1/ Derate linearly 1.2 W/°C for T_A > +25°C. $P_T = \frac{T_J \text{ max} - T_C}{R_{\theta JA}}$

2/ $I_D = \frac{T_J(\text{max}) - T_C}{R_{\theta JA} \; x \; (R_{DS(on)} \; at \; T_J(\text{max})}$

3/ I_{DM} = 4 I_{D1}; I_{D1} as calculated in footnote 2/.

1.4 Primary electrical characteristics at T_C = +25°C.

	Min $V_{(BR)DSS}$		$V_{GS(th)1}$		Max I_{DSS1}		Max $R_{DS(on)1}$ 1/	
	N-channel	P-channel	N-channel	P-channel	N-channel	P-channel	N-channel	P-channel
Type 2N7337	V_{GS} = 0 V I_D = 1.0 mA	V_{GS} = 0 V I_D = -1.0 mA	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA	$V_{DS} \geq V_{GS}$ I_D = -0.25 mA	V_{DS} = 80 percent of rated V_{DS}		V_{GS} = 10 V I_D = I_{D2}	V_{GS} = -10 V I_D = I_{D2}
	V dc		V dc		µA dc		ohms	
	100	-100	min max 2.0 4.0	min max -2.0 -4.0	25	-25	0.7	0.7

1/ Pulsed (see 4.5.1).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

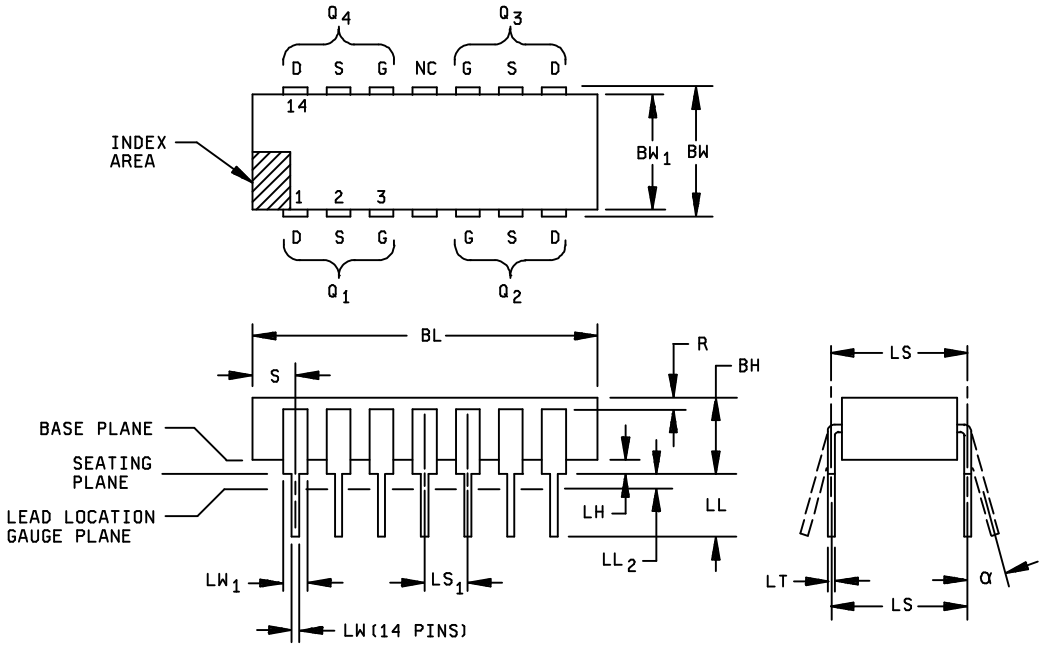
3.3 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (MO-036AB) herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-STD-750, MIL-PRF-19500, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection.

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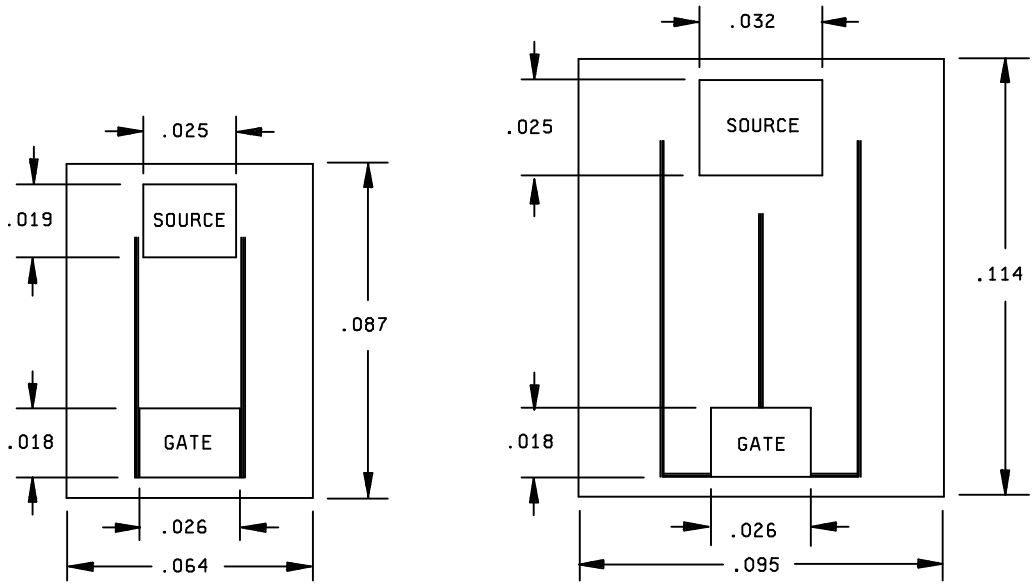


Symbol	Dimension			
	Inches		Millimeters	
	Min	Max	Min	Max
BH	.105	.165	2.67	4.19
LH	.025	.050	0.64	1.27
LW	.015	.021	0.381	0.533
LW_1	.042	.058	1.07	1.47
LT	.008	.012	0.203	0.305
BL	.735	.760	18.67	19.30
BW	.295	.322	7.49	8.18
BW_1	.280	.305	7.11	7.75

Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
LS	.290	.315	7.37	8.00
LS_1	.095	.105	2.41	2.67
LL	.125	.175	3.18	4.45
LL_1	.000	.030	0.00	0.76
α	0°	15°	0°	15°
R	.010	- - -	0.25	- - -
S	.030	.095	0.76	2.41

FIGURE 1. Dimensions and configuration (MO-036AB).

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HEX-1: 100V, N-CHANNEL

HEX-2: 100V, P-CHANNEL

Inches	mm	Inches	mm
.018	0.46	.064	1.63
.019	0.48	.087	2.21
.025	0.64	.095	2.41
.026	0.67	.114	2.90
.032	0.82		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm).
4. The physical characteristics of the die thickness are .0187 (0.474 mm). The metals are chromium, nickel, and silver. The top metal is aluminum and the back contact is the drain.

FIGURE 2. JANC A die dimensions.

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3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of electrostatic charge. The following handling practices shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100$ k, whenever bias voltage is to be applied drain to source.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3 herein.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-PRF-19500 and table III herein.

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4.3 Screening (JANS, JANTX and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table IV of MIL-PRF-19500	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Gate stress test (see 4.5.5)	Gate stress test (see 4.5.5)
1/	$Z_{\theta JC}$ Method 3161 (see 4.5.3)	$Z_{\theta JC}$ Method 3161 (see 4.5.3)
9 1/	I_{GSS1}, I_{DSS1}, subgroup 2 of table I herein	Subgroup 2 of table I herein
10	Method 1042, test condition B	Method 1042, test condition B
11	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$ Subgroup 2 of table I herein ΔI_{GSS1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 25 µA dc or ± 100 percent of initial value, whichever is greater.	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$ Subgroup 2 of table I herein
12	Method 1042, test condition A	Method 1042, test condition A, optional t = 48 hours at 175°C
13	Subgroups 2 and 3 of table I herein ΔI_{GSS1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 25 µA dc or ± 100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ± 20 percent of initial value $\Delta V_{GS(th)1}$ = ± 20 percent of initial value	Subgroup 2 of table I herein ΔI_{GSS1} = ± 20 nA dc or ± 100 percent of initial value, whichever is greater. ΔI_{DSS1} = ± 25 µA dc or ± 100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ± 20 percent of initial value $\Delta V_{GS(th)1}$ = ± 20 percent of initial value

1/ Shall be performed anytime before screen 10.

4.3.1 Screening (JANC). Screening shall be in accordance with appendix E, table IV ofMIL-PRF-19500.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. Alternate flow is allowed for quality conformance inspection in accordance with appendix E of MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. (End-point electrical measurements shall be in accordance with the applicable steps of table III herein.)

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-PRF-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.2.1 Group B inspection, appendix E, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	1051	Test condtion G, 100 cycles.
B4		Not applicable.
B5	1027	Condition A: V_{DS} = rated, T_A = +175°C, t = 120 hours. Read and record $V_{BR(DSS)}$ (pre and post) at I_D = 1.0 mA (N-CH), I_D = -1.0 mA (P-CH). Read and record I_{DSS} (pre and post), in accordance with table III. (NOTE: A separate sample may be pulled for each test)
B5	1042	Condition B: V_{GS} = rated, T_A = +175°C, t = 24 hours. (NOTE: A separate sample may be pulled for each test)
B6	3161	See 4.5.2.

4.4.2.2 Group B inspection, appendix E, table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Conditions
B2	1051	Test condtion G, 25 cycles.
B3	1027	Test condition A: V_{DS} = 80 percent of rated. T_A = +150°C, t = 160 hours. (NOTE: A separate sample may be pulled for each test)
B3	1042	Test condition B: V_{GS} = 80 percent of rated. T_A = +150°C, t = 24 hours. (NOTE: A separate sample may be pulled for each test)
B5 and B6		Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.3.1 Group C inspection, appendix E, table VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	1056	Test condition E; The sampling plan applies to the number of leads tested. A minimum of three devices shall be tested.
C6	1026	Test condition A: V_{DS} = 80 percent of rated. T_A = +150°C, t = 340 hours.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. $R_{\theta JA}$ maximum = 90°C/W, for each die.

a. I_M measuring current ..... 10 mA

b. I_H drain heating current ..... 0.15 A minimum.

c. t_H heating time ..... Steady-state (see method 3161 of MIL-STD-750 for definition)

d. V_H drain-source heating voltage ..... 15 V

e. t_{MD} measurement time delay ..... 30 to 60 μs

f. t_{SW} sample window time ..... 10 μs maximum

4.5.3 Thermal response ($Z_{\theta JC}$ measurements). The $Z_{\theta JC}$ measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit for $Z_{\theta JC}$ in screening (appendix E, table IV of MIL-PRF-19500) shall be derived by each vendor by means of Statistical Process Control (SPC). When three (minimum) lot date codes have exhibited control, the data from these three minimum lot date codes will be used to establish a fixed screening limit, (not to exceed the group A subgroup 2 limit). Once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition. This procedure may be used in lieu of an in line process monitor. The following parameter measurements shall apply:

a. I_M measuring current ..... 10 mA

b. I_H drain heating current ..... 0.15 A minimum.

c. t_H heating time ..... 100 ms.

d. V_H drain-source heating voltage ..... 15 V

e. t_{MD} measurement time delay ..... 30 to 60 μs

f. t_{SW} sample window time ..... 10 μs maximum

4.5.4 Single pulse avalanche energy E_{AS}

a. Peak current I_{AS} ..... I_{D1}

b. Peak gate voltage V_{GS} ..... 10 V

c. Gate to source resistor R_{GS} ..... $25 \leq R_{GS} \leq 200\Omega$

d. Initial case temperature ..... +25°C +10°C, -5°C

e. Inductance ..... $\left[\frac{2E_{AS}}{(I_{D1})^2}\right]$ $\left[\frac{(V_{BR} - V_{DD})}{V_{BR}}\right]$ mH minimum

f. Number of pulses to be applied ..... 1 pulse minimum

g. Supply voltage V_{DD} ..... 25 V (N-channel); -25 V (P-channel)

4.5.5 Gate stress test.

a. V_{GS} = 30 V minimum.

b. t = 250 μs minimum.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal response	3101	See 4.5.3	$Z_{\theta JC}$			
Breakdown voltage, drain to source	3407	V_{GS} = 0 V, condition C	$V_{(BR)DSS}$			
N-channel		I_D = 1.0 mA dc		100		V dc
P-channel		I_D = -1.0 mA dc		-100		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$	$V_{GS(th)1}$			
N-channel		I_D = 0.25 mA dc		2.0	4.0	V dc
P-channel		I_D = -0.25 mA dc		-2.0	-4.0	V dc
Gate current	3411	V_{GS} = +20 V and -20 V, bias condition C, V_{DS} = 0	I_{GSS1}		± 100	nA dc
Drain current	3413	V_{GS} = 0 V; V_{DS} = 80 percent of rated V_{DS}, bias condition C	I_{DSS1}			
N-channel					25	µA dc
P-channel					-25	µA dc
Static drain to source on-state resistance	3421	Pulsed (see 4.5.1), condition A, I_D = rated I_{D2} (see 1.3)	$r_{DS(ON)1}$			
N-channel		V_{GS} = 10 V			0.7	Ω
P-channel		V_{GS} = -10 V			0.7	Ω
Forward voltage (source drain diode)	4011	Pulsed (see 4.5.1), V_{GS} = 0 V, I_D = rated I_{D1}	V_{SD}			
N-channel		V_{GS} = 10 V			1.5	V dc
P-channel		V_{GS} = -10 V			-5.5	V dc
Subgroup 3						
High temperature operation:		$T_C = T_J$ = +125°C				
Gate current		Bias condition C, V_{GS} = +20 V and -20 V; V_{DS} = 0 V	I_{GSS2}		± 200	nA dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Drain current	3413	Bias condition C, V_{GS} = 0 V				
N-channel		V_{DS} = 100 percent of rated V_{DS}	I_{DSS2}		1.0	mA dc
		V_{DS} = 80 percent of rated V_{DS}	I_{DSS3}		0.25	mA dc
P-channel		V_{DS} = 100 percent of rated V_{DS}	I_{DSS2}		-1.0	mA dc
		V_{DS} = 80 percent of rated V_{DS}	I_{DSS3}		-0.25	mA dc
Static drain to source on-state resistance	3421	Pulsed (see 4.5.1); I_D = rated I_{D1}	$r_{DS(ON)3}$			
N-channel		V_{GS} = 10 V			1.4	Ω
P-channel		V_{GS} = -10 V			1.4	Ω
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$	$V_{GS(th)2}$			
		I_D = 0.25 mA dc				
N-channel		I_D = -0.25 mA dc		1.0		V dc
P-channel				-1.0		V dc
Low temperature operation:		$T_C = T_J$ = -55°C				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$	$V_{GS(th)3}$			
		I_D = 0.25 mA dc				
N-channel		I_D = -0.25 mA dc			5.0	V dc
P-channel					-5.0	V dc
Subgroup 4						
Switching time test	3472	I_D = rated I_{D1}; R_G = 24Ω (N-CH); R_G = 18Ω (P-CH) V_{DD} = 0.5 $V_{BR(DSS)}$				
Turn-on delay time		V_{GS} = 10 V	$t_{d(ON)}$			
		V_{GS} = -10 V				
N-channel					20	nS
P-channel					30	nS
Rise time		V_{GS} = 10 V	t_r			
		V_{GS} = -10 V				
N-channel					25	nS
P-channel					60	nS

See footnote at end of table.

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TABLE I. Group A inspection - Continued

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Turn-off delay time			$t_{d(OFF)}$			
N-channel		V_{GS} = 10 V			40	nS
P-channel		V_{GS} = -10 V			60	nS
Fall time			t_r			
N-channel		V_{GS} = 10 V			40	nS
P-channel		V_{GS} = -10 V			60	nS
Subgroup 5						
Single pulse avalanche energy	3470	See 4.5.4	E_{AS}			
Electrical measurements		See table III herein; steps 1, 2, 3, 4, 5 and 6				
Safe operating area test	3474	See figure 4, V_{DS} = 80 percent of rated $V_{BR(DSS)}$, tp = 10 ms				
N-channel		I_D = 0.25 A				
P-channel		I_D = -0.25 A				
Electrical measurements		See table III herein, steps 1, 2, 3, 4, 5, 6, and 7				
Subgroup 6						
Not applicable						
Subgroup 7						
On state gate charge	3471	Condition B	$Q_{g(on)}$			
N-channel					15	nC dc
P-channel					22	nC dc
Gate to source charge			Q_{gs}			
N-channel					7.5	nC dc
P-channel					8.0	nC dc
Gate to drain charge			Q_{gd}			
N-channel					7.5	nC dc
P-channel					14.0	nC dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7 - Continued						
Reverse recovery time	3473	di/dt ≤ 100 A/μS, I_D = rated I_{D1}	t_{rr}			
		V_{DD} ≤ 30 V				
N-channel					200	nS
P-channel		V_{DD} ≤ -30 V			200	nS

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750	
	Method	Conditions
Subgroup 1		
Temperature cycle	1051	Test condition G, 500 cycles
Hermetic seal		
Fine leak Gross leak		Test conditions G or H Test conditions C or D
Electrical measurements		See table III herein, steps, 1, 2, 3, 4, 5, 6, and 7
Subgroup 2		
Steady-state reverse bias	1042	Condition A, 1,000 hours
Electrical measurements		See table III herein, steps 1, 2, 3, 4, 5, and 6
Steady-state gate bias	1042	Condition B, 1,000 hours
Electrical measurements		See table III herein, steps 1, 2, 3, 4, 5, and 6
Subgroup 3		
Destructive physical analysis (DPA)	2102	
Subgroup 4		
Thermal resistance	3161	$R_{\theta JA1}$ = 90°C/W maximum, see 4.5.2
Subgroup 5		
Not applicable		

1/ For sampling plan, see MIL-PRF-19500.

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TABLE III. Groups A, B, C, and E electrical end-point measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Breakdown voltage drain to source	3407	V_{GS} = 0 V, bias condition C	$V_{(BR)DSS}$			
	N-channel		I_D = 1.0 mA dc		100		V dc
	P-channel		I_D = -1.0 mA dc		-100		V dc
2.	Gate to source voltage (threshold)	3404	$V_{DS} \geq V_{GS}$,	$V_{GS(th)1}$			
	N-channel		I_D = 0.25 mA dc		2.0	4.0	V dc
	P-channel		I_D = -0.25 mA dc		-2.0	-4.0	V dc
3.	Gate current	3411	V_{GS} = ± 20 V dc, bias condition C	I_{GSS1}		± 100	nA dc
4.	Drain current	3413	V_{GS} = 0, V_{DS} = 80 percent of rated V_{GS} , bias condition C	I_{DSS1}			
	N-channel					25	µA dc
	P-channel					-25	µA dc
5.	Static drain to source on-state resistance	3421	Condition A, pulsed (see 4.5.1), $I_D = I_{D2}$	$r_{DS(on)1}$			
	N-channel		V_{GS} = 10 V			0.7	Ω
	P-channel		V_{GS} = -10 V			0.7	Ω
6.	Forward voltage (source-drain diode)	4011	Pulsed (See 4.5.1), V_G = 0 V, $I_D = I_{D1}$	V_{SD}			
	N-channel					1.5	V dc
	P-channel					-5.5	V dc
7.	Thermal response	3161	See 4.5.3	$Z_{\theta JC}$			

See footnotes at end of table.

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TABLE III. Groups A, B, C, and E electrical end-point measurements - Continued.

1/ The electrical measurements for appendix E, table VIa (JANS) of MIL-PRF-19500 are as follows:

a. Subgroup 3, see table III herein, steps 1, 2, 3, 4, 5, 6, and 7.

b. Subgroup 5, see table III herein, steps 1, 2, 3, 4, 5, and 6. (No more than 15 percent of the sample shall be permitted to have a $\Delta V_{BR(DSS)}$ shift of more than 10 percent and ΔI_{DSS} greater than 50 μA.).

2/ The electrical measurements for appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1, 2, 3, 4, 5, 6, and 7.

b. Subgroup 3, see table III herein, steps 1, 2, 3, 4, 5, and 6.

3/ The electrical measurements for appendix E, table VII of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1, 2, 3, 4, 5, 6, and 7 for all levels.

b. Subgroup 3 and 6, see table III herein, steps 1, 2, 3, 4, 5, and 6 for all levels.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Substitution information. Devices covered by this specification are substitutable for the manufacturers' and users' Part or Identifying Number (PIN). This information in no way implies that manufacturers' PIN's are suitable as a substitute for the military PIN.

Military PIN	Manufacturer's CAGE code	Manufacturer's and user's PIN
2N7337	59993	IRFG5110

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:)
Army - AR, MI, SM
Navy - AS, CG, MC, OS
Air Force - 13, 19, 85, 99

Preparing activity:
DLA - CC

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/607A	**2. DOCUMENT DATE (YYMMDD)** 980211

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTORS, N-CHANNEL AND P-CHANNEL, SILICON TYPES 2N7337 JANTX, JANTXV, JANS AND JANC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. Point of contact: Alan Barone	b. TELEPHONE Commercial: 614-692-0510 DSN: 850-0510 FAX: 614-692-6939 EMAIL: alan_barone@dscc.dla.mil
c. ADDRESS: Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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